

AC/DC Signal Mixer, Follower, Buffer and Inverter with 10 Inputs



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Often we want to add multiple AC/DC signals to produce the needed composite output signal. One such instance is when we want to add several audio signals for a home entertainment system, or when we wish to add several sinusoidal, triangular or rectangular signals. Another case is when we wish to add several arbitrary AC and DC voltages and produce a composite signal for testing purposes.

Here is a possible solution of a mixer and buffer working in the range of 0Hz to over 100kHz. The circuit can be used as part of audio equipment or as part of equipment for testing and measurement.

Circuit and working

Fig. 1 shows a simple AC/DC mixer with 10 inputs. It is built around dual op-amp NE5532 or RC4560 (IC1) and a few other components. Each input can be used for AC or AC + DC signals. Inputs on connector CON1 (DC1 through DC10) are for AC + DC signals. Inputs on connector CON2 (AC1 through AC10) are for AC only signals.

Any combination of inputs can be used but every input has usage as

PARTS LIST

Semiconductors:

- IC1 - NE5532 or RC4560 op-amp
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R15 - 10-kilo-ohm
- R16 - 2.2-kilo-ohm
- R17 - 100-ohm

Capacitors:

- C1-C10 - 0.1 μ F ceramic disk
- C11, C12 - 0.33 μ F ceramic disk
- C13, C14 - 220 μ F, 35V electrolytic
- C15 - 47 μ F, 35V electrolytic

Miscellaneous:

- CON1, CON2 - 20-pin, 2-line female connector
- CON3 - 3-pin connector
- CON4-CON6 - 2-pin connector
- ± 9 V DC power supply

AC only or as AC + DC input. All inputs have the same parameters and the gain of all inputs is unity.

Input resistance of all inputs is 10-kilo-ohm and these can be driven by ordinary operational amplifiers (OAs) and most signal sources without any problem. Resistors R1 through R10 can be changed to any appropriate value.

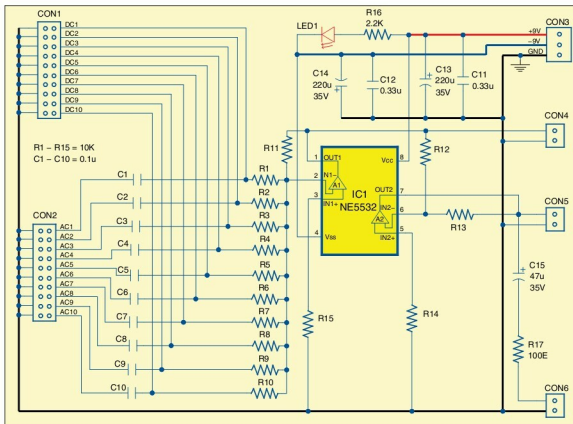


Fig. 1: Circuit diagram of the AC/DC signal mixer, follower, buffer and inverter with 10 inputs

Integrated circuit RC4560/NE5532 has two operational amplifiers. The first op-amp A1 of IC1 is used as a summing and inverting amplifier for all inputs. The resulting signal that is the inverted sum of all input signals with the same weight in the sum is presented on output connector CON4. This signal is inverted by the second op-amp A2 of IC1 and the result is available on connector CON5.

This way, the